

REYHANEH VOJOUDI SALMANI

M.Sc. Physics · Gravity & Cosmology | AI-Driven Astrophysics

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RESEARCH INTERESTS

Machine Learning & AI for Astrophysics · Black Hole Physics & AGN · High-Redshift Galaxies & Quasars · Galaxy Formation & Evolution · Gravitational Theory & Cosmology

EDUCATION

M.Sc. in Gravity and Cosmology

Sep 2020 – Sep 2022

K. N. Toosi University of Technology, Tehran, Iran (Top 5 university in Iran)

First Class Honours — Top Student

- Thesis: "Primordial Black Holes as Dark Matter" · Grade: 20/20 (Excellent)
- Supervisor: Dr. Javad Taghizadeh Firouzjaee

B.Sc. in Physics

Sep 2016 – Sep 2020

K. N. Toosi University of Technology, Tehran, Iran

- Thesis: "Analogue Black Holes"
- Supervisor: Dr. Javad Taghizadeh Firouzjaee

PUBLICATIONS

Manuscript in Preparation

- R. V. Salmani, M. Berahman, and S. Ansarifard. "Foundation Models for Bright Quasar Selection in the Euclid Deep Field South." [In prep.]

Peer-Reviewed & Submitted

- S. Fakhry, R. V. Salmani, "Poisson Fluctuations from Supermassive Primordial Black Holes in Mass-Energy Cascade Models: Implications for High-redshift JWST Galaxies" In preparation. Submitted to Physical Review D, June 2026. [Submitted]
- P. Khaliliyan, R. V. Salmani, J. T. Firouzjaee. "Supermassive Black Hole Mass Evolution as a Stochastic Markov Process: A Unified Machine-Learning and Analytical Framework for High-Redshift Quasars." Submitted to Physical Review D, June 2026. [Submitted]
- S. Fakhry, R. V. Salmani, J. T. Firouzjaee. "High-redshift Galaxies from Early JWST Observations in More Realistic Dark Matter Halo Models." Physical Review D 112, 123503, December 2025. [click]
- S. S. Tabasi, R. V. Salmani, P. Khaliliyan, J. T. Firouzjaee. "Modeling the Central Supermassive Black Hole Mass of Quasars via LSTM Approach." The Astrophysical Journal, September 2023. [click]

RESEARCH EXPERIENCE

Research Assistant — Institute for Research in Fundamental Sciences (IPM)

Oct 2025 – Present

Supervisors: Dr. Saeed Ansarifard & Dr. Michael Fausnaugh · Tehran, Iran & Texas, USA

- Developing and fine-tuning transformer-based foundation models for quasar identification in the Euclid Deep Fields.
- Applying self-supervised learning with masked autoencoders to learn representations of astronomical data.
- Contributing to a research manuscript in preparation for journal submission

Research Assistant — K. N. Toosi University of Technology

May 2024 – Present

Supervisors: Dr. Saeed Fakhry & Dr. Javad Taghizadeh Firouzjaee · Tehran, Iran

- Analyzed high-redshift galaxy candidates from early JWST data within extended dark matter halo models.
- Built data-driven LSTM and Markov-chain models for supermassive black hole mass evolution in high-z quasars (submitted Phys. Rev. D 2026)

Research Collaborator — Spherical Domain Wall Collapse

Jan 2023 – Apr 2024

Supervisor: Dr. Nima Khosravi · Tehran, Iran

- Investigated dynamics of spherical domain wall collapse within gravitational theories

Research Assistant — Physics Data and Astronomy Technology Lab (PDAT)

Apr 2021 – Sep 2023

Supervisor: Dr. Javad Taghizadeh Firouzjaee · Tehran, Iran

- Studied AGN, supermassive black holes, and primordial black holes
- Contributed to publications in Phys. Rev. D and The Astrophysical Journal

TECHNICAL SKILLS

Programming: Python (ML/AI, data analysis, scientific computing), C++, Wolfram Mathematica, LaTeX
ML Methods: Autoencoders, LSTM, Markov-chain, applied to astrophysical datasets
Cosmology Software: CosmoMC, CAMB (workshop training)
Languages: Persian (Native) · **English C1 (TOEFL 108/120)** · **German C1** · French / Italian / Arabic

HONOURS & AWARDS

Fully-Funded Ph.D. Offer — Virginia Tech University, USA Fall 2025
Supervisor: Prof. Nahum Arav

Fully-Funded Ph.D. Offer — Texas Tech University, USA Fall 2025
Supervisor: Prof. Michael Fausnaugh

M.Sc. First Class Honours — Top Student Award 2022
K. N. Toosi University of Technology

Silver Medal — Persian Young Physicist's Tournament (PYPT) 2024
Mentored a team of 6 students; 4 selected for Iran's national team

Bronze Medal & National Team Selection — PYPT 2014
Selected as a qualified member of Iran's national team

TEACHING EXPERIENCE

Teaching Certification — Texas Tech University, USA August 2025
Approved to teach in English-language university classrooms (Valid 2025–2030)

Teaching Assistant — General Relativity (M.Sc. level) Sep 2023 – Jan 2024
K. N. Toosi University of Technology · Lecturer: Prof. Hadi Hedayati

Teaching Assistant — Cosmology (M.Sc. level) Sep 2020 – Jan 2021
K. N. Toosi University of Technology · Lecturer: Prof. M. H. Z. Haghighi

Head of Education — Implementing AI in International Education Jan 2024 – April 2025
Iranian Cambridge School - ICS · Tehran, Iran

Lecturer — IGCSE/A-Level Physics & Mathematics (English instruction) Sep 2020 – Present
Various institutions, including Glocal College, Hakim International School, and Nikoo International School · Tehran, Iran

WORKSHOPS, SCHOOLS & CONFERENCES

AI-Driven High Energy Physics Graduate Course (92.5/100) Oct 2025 – Jan 2026
Institute for Research in Fundamental Sciences (IPM) · Tehran, Iran

Python Programming Course (100/100) Sep 2025
Scientific Association of Robotics and Mechatronics at Alzahra University · Tehran, Iran

Workshop on High Energy Physics Feb 2024
Institute for Research in Fundamental Sciences (IPM) · Tehran, Iran

Oral Presentation — Annual Physics Conference of Iran Sep 2023
University of Isfahan · Topic: "Modeling Quasars' Central Supermassive Black Holes"

High Energy Physics School Nov 2023
Institute for Research in Fundamental Sciences (IPM) · Tehran, Iran

Workshop on Cosmology: From Theory to Observation Aug 2023
Institute for Research in Fundamental Sciences (IPM) · Tehran, Iran

Cosmology with New Data Conference (Executive Committee Member) July 2021
K. N. Toosi University of Technology · Tehran, Iran

REFERENCES

Dr. Javad Taghizadeh Firouzjaee, Associate Professor, K. N. Toosi University of Technology · firouzjaee@kntu.ac.ir
Dr. Saeed Fakhry, Postdoctoral Researcher, University of Valencia · saeed.fakhry@uv.es
Dr. Majid Vaezzadeh, Professor, K. N. Toosi University of Technology · majid@kntu.ac.ir